
LF ENERGY
Technical Advisory Council (TAC) Meeting
October 13, 2026

Meeting information

- ➔ Meeting to begin at 5:00 pm Central European Time
- ➔ Join the meeting at the link in your calendar in LFX Individual Dashboard
- ➔ Any problems with connectivity, you can contact John Mertic from the Linux Foundation at +1 234-738-4571 or Yarille Ortiz at +1 585-967-3585
- ➔ Previous TAC Meeting notes, deck, and recording, at https://tac.lfenergy.org/meetings/

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Agenda

All Times in Central European Time Zone

- 5:00 pm - 5:15 pm - Opening and General Updates
  - TAC member updates and annual review date reminders
  - Project Pipeline
  - TAC Chair/Vice Chair Election Update
  - Project AI Guidelines #864
- 5:15 pm - 5:25 pm - OneNet Framework Follow-Up #112
- 5:25 pm - 5:40 pm - Annual Review - SOGNO #75
- 5:40 pm - 5:55 pm - Annual Review- URPX #600
- 5:55 pm - 6:10 pm - Marketing and PR Updates
- 6:10 pm - 6:15 pm - Closing and Next Meeting

Opening and General Updates

5:00 pm – 5:15 pm

Technical Advisory Council (TAC)



Antonello Monti

Chair

Professor

RWTH Aachen University



Art Pope

Member of Technical Staff

Google LLC



Boris DOLLEY

Director of OSPO and Sustainable IT Strategy

RTE (Reseau de Transport dElectricite)



Frédéric Didier

Lead tech

RTE (Reseau de Transport dElectricite)



Jonas van den Bogaard

Vice Chair

Open Source Office Lead

Alliander



Maarten Mulder

PO IoT Field Device Platforms

Alliander



Moïse K. Kameni

Entreprise Architect and Head of Open Source Program Office

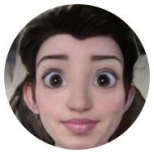
Hydro-Québec



Sachin Bhakar

Strategy Advisor - Computational Science & Digital Innovation

Shell Energy Retail Limited



Sophie Frasnedo

Software developer

RTE (Reseau de Transport dElectricite)



Travis Sikes

Data Science Manager

Recurve

LFENERGY

Projects

LF ENERGY



TAC Meeting Schedule 2026

The TAC meetings are monthly, on the second Tuesday of the month at 8:00am US Pacific Time/11:00am US Eastern Time unless otherwise noted.

- ~~January 13~~
- ~~February 10~~
- ~~March 10~~
- ~~April 14~~
- ~~May 12~~
- ~~June 9~~
- ~~July 21 (rescheduled from 14)~~
- ~~*August 11 - cancelled*~~
- ~~*September 8*~~
- ~~*September 14 joint Board and TAC meeting at LF Energy Summit*~~
- ~~October 13~~
- **November 10**
- December 8

Project and Working Group Leads

Name	Chair
AINETUS	Ricardo Bessa
Arras	Alyona Teyber
AssetLife	William Grison
Battery Data Alliance	Gabe Hege
CitrineOS	Thana Paris
CityLearn	Zoltan Nagy
CoMPAS	Sander Jansen
Connected Data Specification - Customer Data Working Group (CDS WG3)	Daniel Roesler
Connected Data Specification - Power Systems Data Working Group (CDS WG2)	Stephen Suffian
Connected Data Specification - Registration Working Group (CDS WG1)	Daniel Roesler
covXtreme	Sachin Bhakar
CUPID (Controllable Unit Protocol Interface for DER)	Alexandre Lucas
Dynaωo	Marco Chiaramello
EnerGNN	Balthazar Donon
EVerest	Marco Möller
FIDOPower	Alyona Teyber
FlexMeasures	Nicolas Höning
Grid Edge Interoperability & Security Alliance (GEISA)	Michael Stuber, Richard Lam
Grid Vantage	Alyona Teyber
Grid2Op	Benjamin Donnot
Hyphae	Arila Barnes
LF Energy RegistryOS	Casey Martinez
LF Energy Semantic Energy Framework (LFE-SEF)	Barry Nouwt
NODE Collective	Deandrea Salvador
OpenDSM	Travis Sikes
OpenGridFM	François Mirallès
OpenLEADR	Hugo Van De Pol
OpenSTEF	Daan Van Es
OpenSynth	Gus Chadney
OperatorFabric	Frédéric Didier
ORES (Open Renewal Energy Systems)	Chris Xie
Power Grid Model	Peter Salemink
Power Stability Wide Area Monitoring Protection (p-SWAMP)	Kjell Petter Myhren
PowSyBL	Sophie Frasnedo
Real Time Data Ingestion Platform (RTDIP)	Chloe Ching
RTC-Tools	Jesús Andrés Rodríguez Sarasty
SEAPATH	Eloi Bail
Shapeshifter	Robben Riksen
Smart HEMS Benchmark	Xiaoke Yang
SOGNO	Antonello Monti
TROLIE	Christopher Atkins
Utility Rate Plan Exchange	Klaartje De Schepper
VAREN	Guillaume Houle

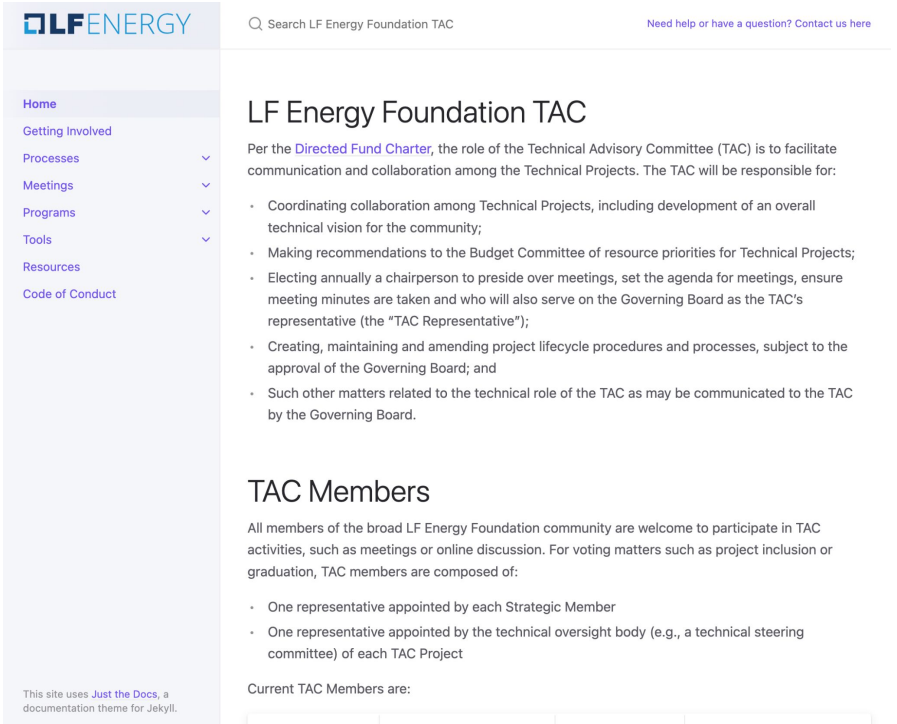
SIGs and SIG Leaders

Name	Chair
AI SIG	Alex Thornton
Edge Interoperability and Flexibility SIG	Robert De Leeuw, Thana Paris
Grid Simulation and Modeling SIG	Thomas Van Dijk
OSPO SIG	Moïse Kameni

TAC Resources

- TAC Website - https://tac.lfenergy.org
  - Contains all the TAC policies and meeting materials, as well as guides to using the various LF Energy tools
- Project Resource Request form - https://tac.lfenergy.org/tools/resource_request.html.

Questions/feedback - let us know!



Annual Review Schedule - TAC

Source:
https://tac.lfenergy.org/process/review_cycle.html

LFENERGY

Name	Last Review Date	Next Review Date
CUPID (Controllable Unit Protocol Interface for DER)	5/13/2025	TBD
Real Time Data Ingestion Platform (RTDIP)	7/8/2025	TBD
~~*SOGNO*~~	~~*5/13/2025*~~	~~*10/13/2026*~~
~~*Utility Rate Plan Exchange*~~		~~*10/13/2026*~~
AI SIG	**11/11/2025**	**11/10/2026**
Grid Edge Interoperability & Security Alliance (GEISA)	**10/14/2025**	**11/10/2026**
RTC-Tools		**11/10/2026**
Connected Data Specification - Customer Data Working Group (CDS WG3)	12/9/2025	12/8/2026
Connected Data Specification - Power Systems Data Working Group (CDS WG2)	12/9/2025	12/8/2026
Connected Data Specification - Registration Working Group (CDS WG1)	12/9/2025	12/8/2026
NODE Collective	12/9/2025	12/8/2026
CityLearn	1/13/2026	1/12/2027
TROLIE	1/13/2026	1/12/2027
Edge Interoperability and Flexibility SIG	2/10/2026	2/9/2027
Grid Simulation and Modeling SIG	2/10/2026	2/9/2027
LF Energy RegistryOS		2/9/2027
AINETUS		3/9/2027
EnerGNN		4/13/2027
LF Energy Semantic Energy Framework (LFE-SEF)	4/14/2026	4/13/2027
Shapeshifter	4/14/2026	4/13/2027
OpenDSM	5/12/2026	5/11/2027
AssetLife		6/8/2027
CoMPAS	6/9/2026	6/8/2027
Smart HEMS Benchmark		6/8/2027
SEAPATH	7/21/2026	7/13/2027
VAREN		7/13/2027
Battery Data Alliance	9/8/2026	9/14/2027

Annual Review Schedule - SIG

SIG Leaders - please share how recent reviews have went, and let us know if the schedule/alignment is still correct - contact email support@lfenergy.org

Source: *https://tac.lfenergy.org/process/review_cycle.html*

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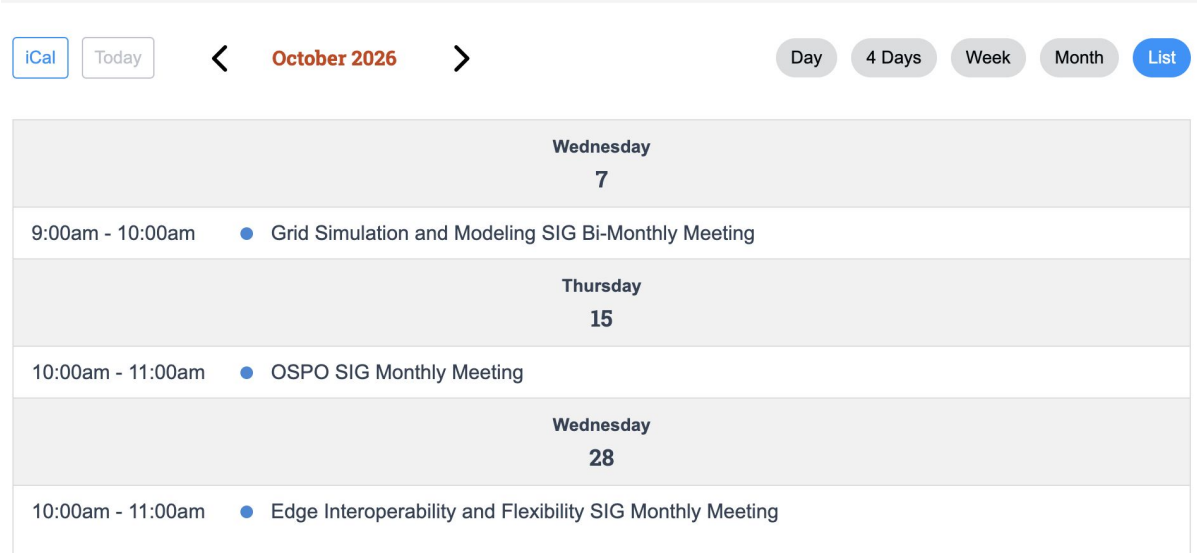
Name	Last Review Date	Next Review Date	SIG
Grid Vantage	9/26/2023	TBD	Grid Simulation and Modeling SIG
covXtreme	4/2/2025	TBD	Grid Simulation and Modeling SIG
Arras	10/1/2025	12/2/2026	Grid Simulation and Modeling SIG
FIDOPower	6/4/2025	12/2/2026	Grid Simulation and Modeling SIG
PowSyBL	12/3/2025	12/2/2026	Grid Simulation and Modeling SIG
CitrineOS	12/17/2025	12/23/2026	Edge Interoperability and Flexibility SIG
FlexMeasures	12/17/2025	12/23/2026	Edge Interoperability and Flexibility SIG
EVerest	1/28/2026	1/27/2027	Edge Interoperability and Flexibility SIG
OpenSTEF	2/4/2026	2/3/2027	Grid Simulation and Modeling SIG
Power Grid Model	2/4/2026	2/3/2027	Grid Simulation and Modeling SIG
Hyphae	3/25/2026	3/24/2027	Edge Interoperability and Flexibility SIG
Dynaωo	4/1/2026	4/7/2027	Grid Simulation and Modeling SIG
OpenSynth	4/21/2026	4/20/2027	AI SIG
ORES (Open Renewal Energy Systems)	4/22/2026	4/21/2027	Edge Interoperability and Flexibility SIG
Grid2Op	6/18/2026	6/17/2027	AI SIG
OpenGridFM	6/18/2026	6/17/2027	AI SIG
OpenLEADR	6/24/2026	6/23/2027	Edge Interoperability and Flexibility SIG

SIG Meeting Schedule for October

All SIG meetings can be found on the LF Energy calendar (calendar.lfenergy.org) as well as the SIG Calendar (sigcalendar.lfenergy.org)

Days/times listed are US Eastern Time

➔ **SIG Leaders – share any updates for your SIGs**



Project Pipeline

https://github.com/orgs/lf-energy/projects/2/views/5

Newly proposed

- TwinREV #937, developed within the Horizon Europe CRETE VALLEY project, is an open source, extensible Digital Twin platform unifying renewable forecasting, power flow simulation, and multi-energy optimization to support data-driven planning and operations.
  - In review by TAC Chair/Vice Chair before proceeding with LF Onboarding.

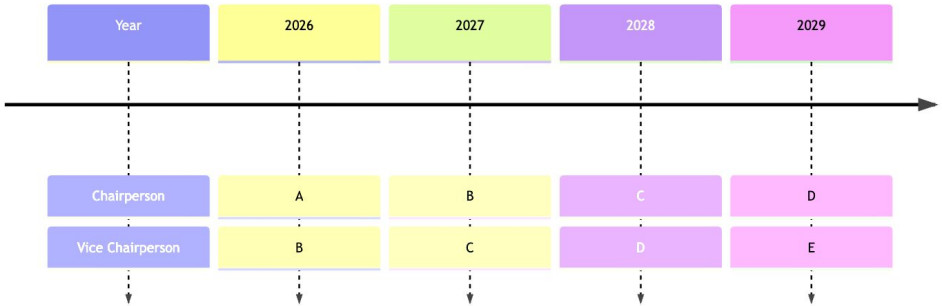
In LF Onboarding

- EDDIE (#719)
  - Waiting on project to clear up trademark issues

Vice Chair Election Update

The TAC Vice Chair election resulted in a tie between nominees Nico Rikken and Boris Dolley.

The candidates collaborated on a transition plan which includes Boris serving for the current term, while Nico begins integrating into TAC planning to ensure a seamless handoff for the following term.



Project AI Guidelines #864

OneNet Framework Follow-Up #112

LFE TAC Meeting – Sept 8th

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Follow up from last TAC meeting

- Presentation from last meeting (https://github.com/lf-energy/tac/blob/main/meetings/2026/2026-09-08/OneNet_Data_Space_for_LFE_v0.2.pdf)
- There were some open questions on project scope.
- For this meeting, need alignment on whether to push ahead for approving this as a project or not.

DISCUSSION: Any open questions/concerns and see if TAC wants to push forward with a vote.

Annual Review: SOGNO #75

5:25 pm – 5:40 pm



Annual Review for SOGNO

SOGNO

Brief Description:

SOGNO creates plug-and-play, cloud-native, micro-services to implement next generation of data-driven monitoring and control systems for grid automation. It simplify the life of utilities by enabling them to optimize their network operations through open source to deliver cost-effectively, and seamless, secure power supply for their customers. SOGNO introduces the idea of grid automation as a modular system in which components can be added through time. This is in opposition to classical monolithic solutions.

TSC Chairperson:

Antonello Monti (amonti@eonerc.rwth-aachen.de)

TSC Members and Affiliations:

Juan Adolfo Galeano, Florian Oppermann, Sanket Gaikwad, Sreejith Pananchickal, Leonardo Carreras *(RWTH Aachen University)*
Varshitha Ramanna, Pranav Kulkarni, Beyza Cizmeci, Varshitha Chamanahalli *(Fraunhofer FIT)*
Markus Mirz (PSI), Roberto Raffo (Google*),* Alberto Patrizi (ARETI)
Holger Blasum (UrStrom Bürgerenergie eG)

Key Links:

GitHub: https://github.com/sogno-platform

Website:

https://www.lfenergy.org/projects/sogno/

Artwork: *N/A*

Mailing lists:

https://lists.lfenergy.org/g/SOGNO-TSC

OpenSSF Best Practice Badge URL:

https://www.bestpractices.dev/en/projects/5627

Organizations/projects contributing/using SOGNO

































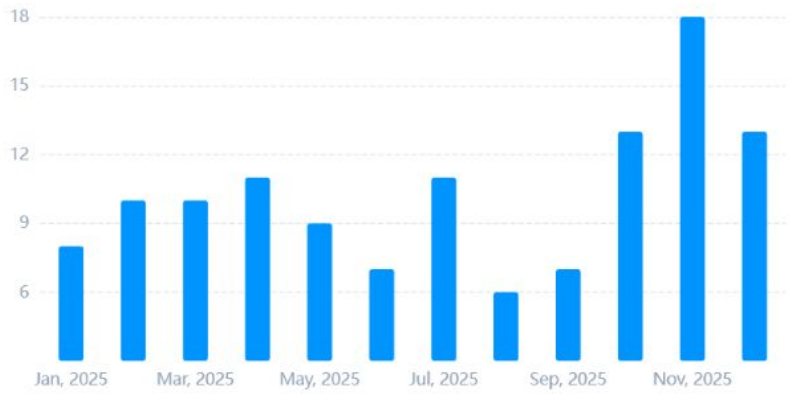
Contributions:

40 25% (+8) vs. 32 last period

Maintainers 8

Reviewers 13

Weekly | Monthly | Quarterly



Organizations leaderboard

Organizations ranked by the number of contribution activities performed by contributors on their behalf during the selected time period. Learn more

Include collaborations

All activities

Organization	Total contributions
RWTH Aachen University	1,587 - 49%
Tom's Café	809 - 25%
OPAL-RT Germany GmbH	481 - 15%
Universidad Nacional de Colombia	153 - 5%
E.ON Energy Research Center	125 - 4%
Loophole Labs Inc	21 - 1%
OZ Sports	17 - 1%
Zaphiro Technologies	10 - 0%
FGH	8 - 0%
Swissgrid AG	5 - 0%

Contributions:

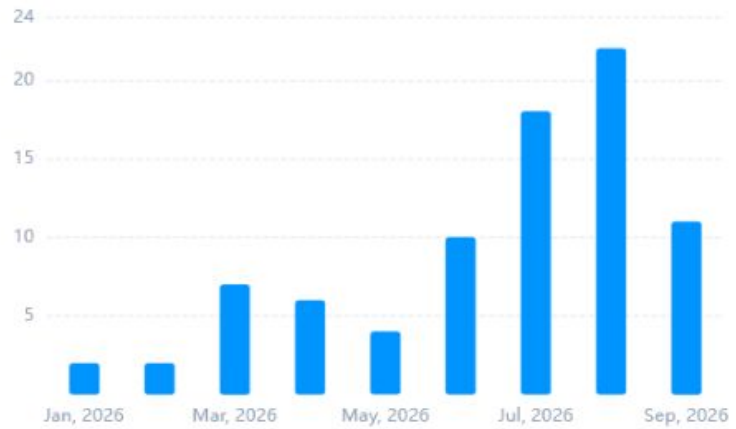
Active contributors

34 3% (+1) vs. 33 last period

Maintainers 0

Reviewers 9

monthly active contributors



Organizations leaderboard

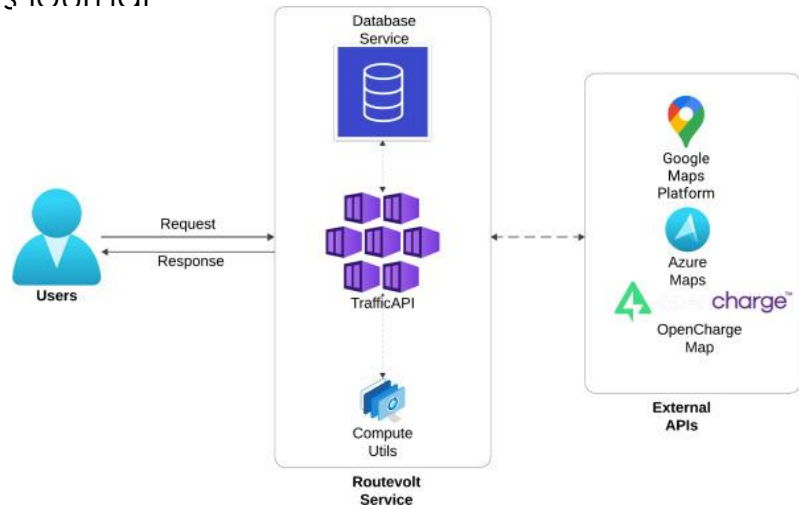
All activities

Organization	Total contributions
RWTH Aachen University	3,015 - 44%
E.ON Energy Research Center	2,030 - 29%
Tom's Café	765 - 11%
OPAL-RT Germany GmbH	430 - 6%
PSI Software AG	379 - 5%
Ita Institut Fur Textiltechnik Der Rwth Aachen Uni	120 - 2%
OPAL-RT TECHNOLOGIES	52 - 1%
Universidad Nacional de Colombia	36 - 1%
FGH	22 - 0%
OZ Sports	17 - 0%

Key Achievements in the past year

Routevolt: New Service

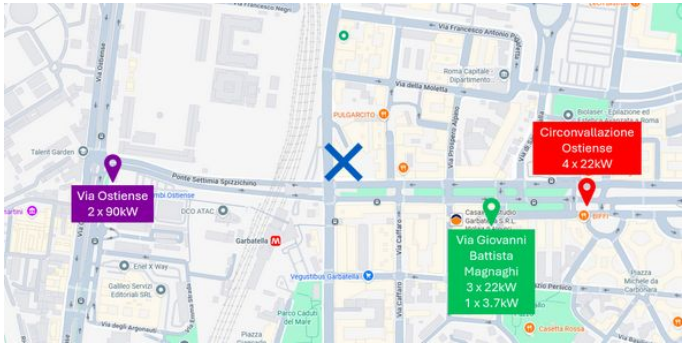
- Is being used as real-time traffic endpoint for evrich. (Advanced estimation of arrival SOC and time calculations to charging points)
- Software paper published in Elsevier Software Impacts journal
- Now available in SOGNO repo.
- DOI: https://doi.org/10.1016/j.simpa.2026.100837

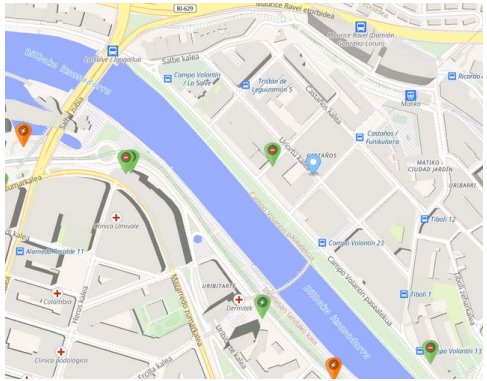


Key Achievements in the past year

Evrich & routevolt: Conducted two demos within two EU projects

- **FLOW: Rome demo**
  - Management of 3 charging stations with total 7 charging ports, helping assess the impact of optimized station selection on reducing charging-related loads by comparing baseline and integrated consumption levels with ARETI.
- **BeFlexible: Bilbao demo**
  - Optimized CP recommendations throughout the 24 h, aiming to improve energy distribution and reduce congestion through a before-and-after analysis of demand patterns with i-de.





Key Achievements in the past year

Pyvolt and PMU based observability demonstrated in the field

BeFlexible: Rome demo

The pyvolt was used with a low cost pmu devoloped by RWTH Aachen. This was used in a secondary substation in Rome with ARETI. The data streamed was used for state estimation, improving the observability and congestion detection.





200 kW Max nominal load

66kW - Average load

Key Achievements in the past year

Proloaf Upgrade



- Improved handling of time features
- Preprocessing steps to create generalizable model (needs evaluation)
- Various QOL improvements
- Integration as Event based Service in OpenEnergyTwin
- Parallel Training now possible

PyMFM Upgrade

- Handling of life battery data
- Various QOL improvements
- Major code restructuring (under review)

Key Achievements in the past year

Evrich Upgrade

- Helm Version (unreleased)
- Improved Simulation allowing Experiment chains and comparison of scenarios (unreleased)

Pipeline Prototype

- Prototype for eventbased architecture.
- Based on Kafka
- Forecasting integrated
- Docker, K8s, Helm
- OIDC under review

Key Achievements in the past year



DPsim

Projects: SEGURO, HYPOWER, CRETE VALLEY

- **New Release: v1.2.1 on Dec 2025**

Available for Linux in PyPI and from source in Github

- **Simulation Models**

State-Space-Nodal component models added, Ph1 and Ph3 line models for Diakoptics

- **Real-Time & VILLAS Enhancements**

Decoupling line EMT models for co-simulation

- **Workflow & CI/CD Upgrades**

Improvements in PyPI publishing, addition of nix build and optimization of container builds.

- **Testing & Examples**

Easy access for user tests with myBinder including VILLASnode support (badge in the landing page of the repository), example Notebooks updated for new models.

- **Code Quality & Maintenance**

Coverage testing added, workflow tests with pre-commit, documentation updates.

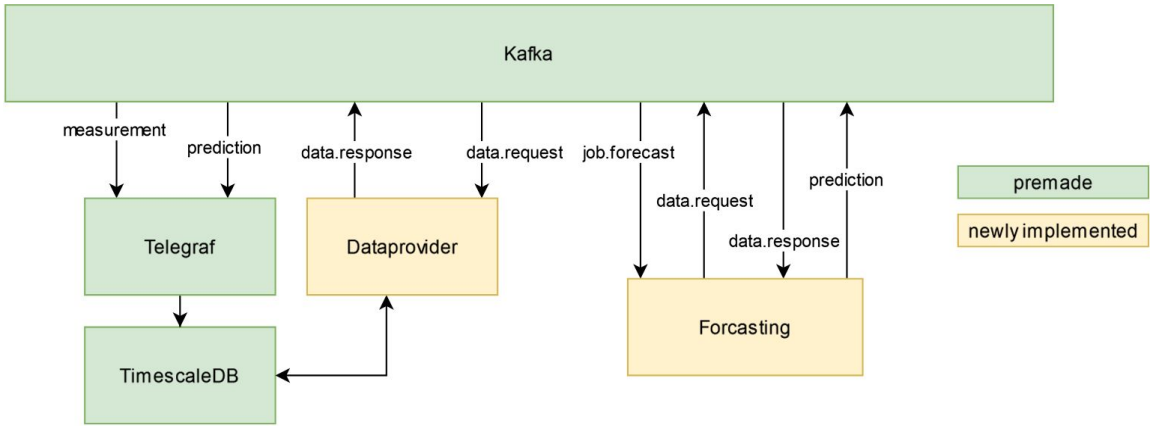
- **Security Aspects**

Trusted publisher added to PyPI and TestPyPI. Progressive dependence upgrades.

Key Achievements in the past year

Continuing adapting the data Architecture (Prototype)

- Prototype
  - Unified way of data and task exchange
  - Expect to tie disparate SOGNO services together
- Workers are always available
- Central data storage
- Both data and tasks come via Kafka
- Services request missing data, Providers fulfil requests autonomously



Prototype architecture and preliminary topics

Key Achievements in the past year

- **InterSCADA** (EU project)
  - Develop an open-source software framework for modular SCADA platform.
    - Checking orchestration mechanism to onboard new microservices. Discussing licensing. There are currently 9 functional microservices.
  - Help grid operators maintain system stability in increasingly hybrid AC/DC power grids
- **OpenEnergyTwin** (OFFIS, RWTH, and Fraunhofer):
  - Integrated Proloaf into OET platform.
  - Advance GUI for visualization of high and low voltage scenario.
  - Project Phase 1 successfully delivered.
  - Preparation for Phase 2.
  - OET results to be used to enhance SOGNO core

Key Achievements in the past year

OpenEnergyTwin (OET) – Phase 1

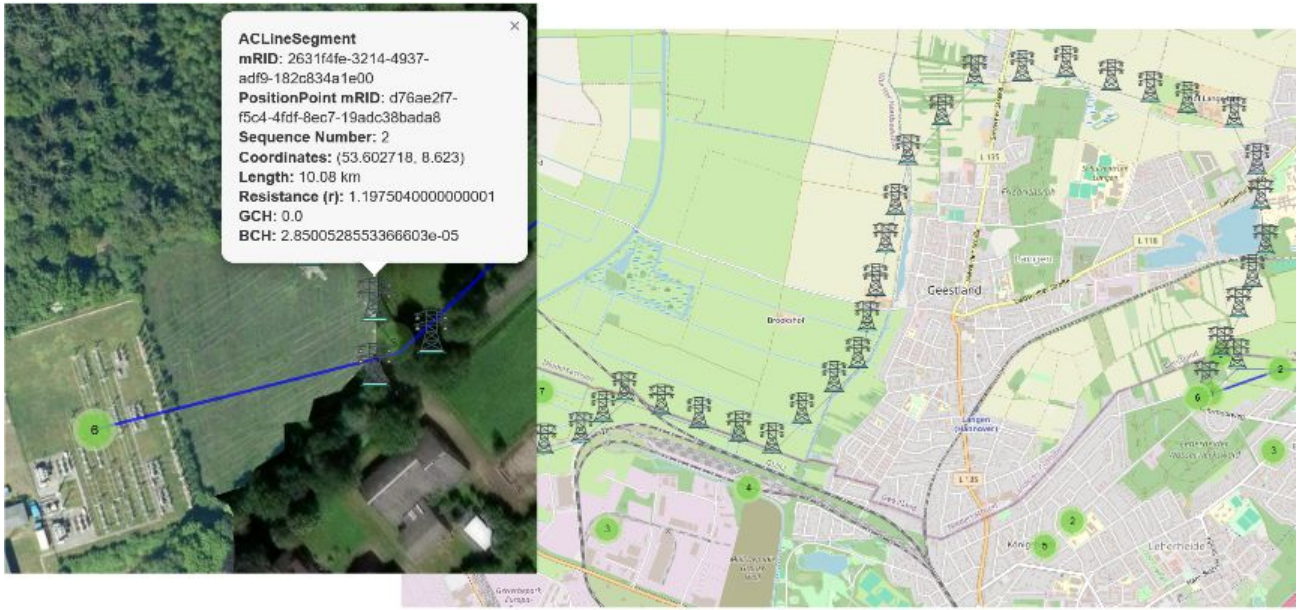
- The human-machine interface auto-drawing topological data and visualizing key live parameter



Key Achievements in the past year

OpenEnergyTwin (OET) – Phase 1

- The human-machine interface auto-drawing topological data and visualizing key live parameter



Key Achievements in the past year

Workshops in ACS and FIT

- A target picture for modular control systems development in Aachen

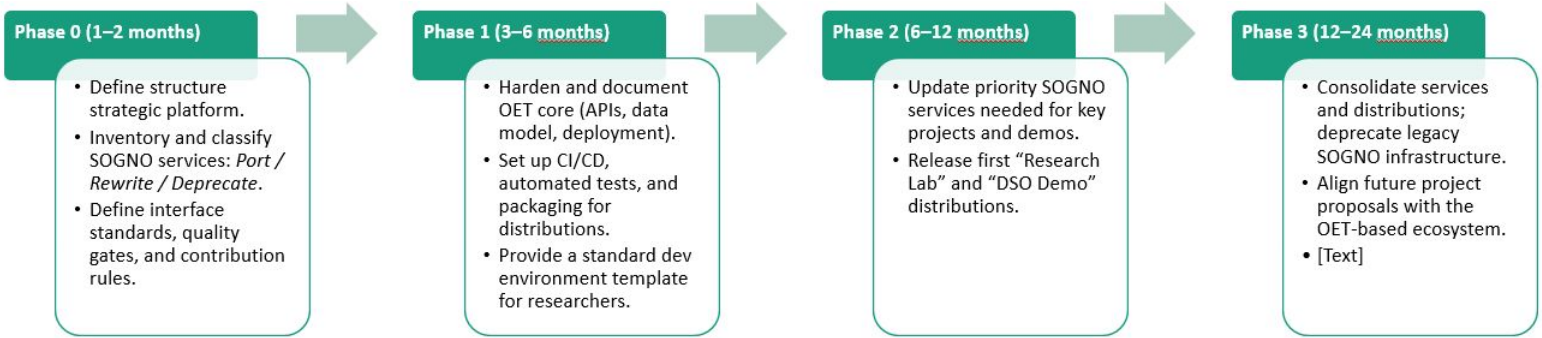
Workshops Objectives

- Understanding core blockers in current SOGNO architecture.
- Reviewing OET architecture for systematic integration and usage of the OET core and GUI.
- Identifying problems related to overlapping platform (OET & SOGNO)
  - Duplicated effort
  - Integration problems
  - Unclear Direction

Key Achievements in the past year

Workshop Results:

- Roadmap and Governance:



Governance:

- Steering group (management + lead engineers): roadmap, major decisions.
- Platform maintainers: OET core, data model, APIs, releases.
- Service owners: specific services (incl. SOGNO-derived), quality and evolution.
- Distribution maintainers: coherence, documentation, and support for each distribution.

Key Achievements in the past year

Workshop Results:

- Alignment on Architecture enhancement
- Results obtained:
  - Expectations from digitalization of SCADA Systems
  - Architecture modules and improvement potentials
  - Suggestions on Governance of Open-Source Software.
  - Established tasks and roadmap align SONGO and OET efforts toward a unified platform.

New key project: OpenAIFlex

- OpenAIFlex is a contract with the European Commission led by INESC-TEC. Fraunhofer is a sub-contractor.
- Thanks to this project some key milestones will be achieved:
  - Integration of the developments made in Rome back in the main line of the project
  - Installation of the SOGNO platform in Vienna and in other DSO in Austria
  - Replica of the experience also in one DSO in South America (negotiation in these days)
  - Creation of a dedicated API for AI-based services
  - Deployment of a first set of AI-driven services in all the active SOGNO locations including Rome

Roadmap 2026

Category	Feature/Improvement/Cooperation	Planned Quarter	Project
New Features	Grid Situational Awareness: Dynamic State Estimator as a micro-service	On Hold	
Improvements	Extend PyVolt to include DC State Estimation	Q2	
			InterScada
			OET
			Seguro
	Use case, requirements, architecture draft	Q2	Joint Sogno Document
	OpenSSF requirements: Best practices	Q3	SOGNO
	OKR Framework for Project management	Running with Phase 0 below	SOGNO
	Security alerts (Dependabot): Proloaf, Pymfm	2026/Q2	
	Complete Evrich REST API	To be defined	
	Set uniform Data Management	Running with Phase 1 below	
	Revise communication schema and make it uniform (DPsim + Pyvolt)	Running with Phase 2 below	
Management Proposed Roadmap	**•Phase0 (1–2 months)**		
	Define structure strategic platform.	2026/Q2	SOGNO
	Inventory and classify SOGNO services: *Port / Rewrite / Deprecate*.	2026/Q2	SOGNO
	Define interface standards, quality gates, and contribution rules.	2026/Q2	SOGNO
	•Phase 1 (3–6 months)		
	Harden and document OET core (APIs, data model, deployment).	2026/Q3	SOGNO
	Set up CI/CD, automated tests, and packaging for distributions.	2026/Q3	SOGNO
	Provide a standard dev environment template for researchers.	2026/Q3	SOGNO
	•Phase 2 (6–12 months)		
	Update priority SOGNO services needed for key projects and demos.	2026/Q4	SOGNO
	Release first “Research Lab” and “DSO Demo” distributions.	2026/Q4	SOGNO
	•Phase 3 (12–24 months)		
	Consolidate services and distributions; deprecate legacy SOGNO infrastructure.	To be defined	SOGNO
	Align future project proposals with the OET-based ecosystem.	To be defined	SOGNO

Growth Plan: Projects

- **CreteValley:** datafev will be utilized to calculate EV flexibility

- **HEDGE-IoT**
  - Utlize *ProLoaF* for demand forecasting as first step of congestion forecasting
  - Update *ProLoaF* with SOTA algorithms(Transformer, TFT)

Areas the project could use help on

- Enhance collaboration with other LF projects
  - Exploring opportunities in the field of Data Spaces (OneNet).
  - Connectors to interact with other platforms in the LFE ecosystem (EVerest).
- Growth opportunities:
  - Promoting existing SOGNO services to potential users.
  - Jointly searching funding opportunities for future developments.
- Guidance on management and quality assessment would be highly appreciated

OpenSSF Best Practices



sogno-platform

Projects that follow the best practices below can voluntarily self-certify and show that they've achieved an Open Source Security Foundation (OpenSSF) best practices badge. Show details

If this is your project, please show your badge status on your project page! The badge status looks like this: openssf best practices in progress 79% Here is how to embed it: Show details

These are the passing level criteria. You can also view the silver or gold level criteria.

Baseline Series: Baseline Level 1 | Baseline Level 2 | Baseline Level 3

Expand panels | Show all details | Show only incomplete criteria

- Basics 13/13
- Change Control 9/9
- Reporting 8/8
- Quality 6/13
- Security 14/16
- Analysis 3/8

Status of the OpenSSF Badge

- OpenSSF's certification is discussed in our regular internal meetings
- SOGNO has several repositories and contributors to align
- Verifying non-compliant repositories and disseminating the requirements among colleagues
- We have more manpower now (RWHT+FIT), and we are refining our workflow
- The Silver Badge is our final target for the whole SOGNO platform

Feedback on working with LF Energy

- Helps to identify common elements that can then be used as a framework to expand the SOGNO suite more cleanly.
- Motivates to produce reusable, open source, and well-documented code during research projects, thus also improving the transfer of learning.
- Identification and alignment with the industry requirements thanks to the open community environment.
- Automated tools are very helpful: License scan and Insights.
- Reviewing the requirements for best practices gives us a new way of working

TAC Open Discussion



Annual Review:
URPX #600

5:40 pm – 5:55 pm

LF ENERGY

Annual Review for Utility Rate Plan Exchange (URPX)

LF Energy TAC, 2026-10-13. Klaartje De Schepper, chair, URPX Working Group

LF ENERGY

Agenda

- URPX: the project and its lifecycle position
- Incubation criteria: a self-assessment
- The past year: contributions and organizations
- How URPX is built and published: repositories, workflow, the ontology
- Key achievements, and where we could use help
- Feedback on working with LF Energy

Utility Rate Plan Exchange (URPX)

Description

Utility Rate Plan Exchange (URPX) aims to develop a comprehensive, standardized method for representing and exchanging utility rate plan data in machine-accessible format.

URPX will provide a solid yet versatile foundation that governments, utilities, vendors, and technology companies can leverage to improve transparency in energy pricing and clean energy procurement.

The project builds upon an existing collaboration between Flux Tailor and the National Laboratory of the Rockies (NLR) and aims to address the fragmented approach currently used by vendors who handle rate plan calculations in silos using custom logic and rules-based systems.

LFESS working group, accepted 2025-11-05. Working Group Draft since 2026-07-17.

Links

Website: https://urpx.org/

Standard: https://github.com/urpx-org/urpx

Working group: https://github.com/urpx-org/urpx-wg

Mailing list: https://lists.lfenergy.org/g/urpx-wg

LF Energy page:
https://lfenergy.org/projects/utility-rate-plan-exchange-urpx/

Meetings:
https://zoom-lfx.platform.linuxfoundation.org/meetings/utility-rate-plan-exchange

Artwork:
https://artwork.lfenergy.org/projects/utility-rate-plan-exchange

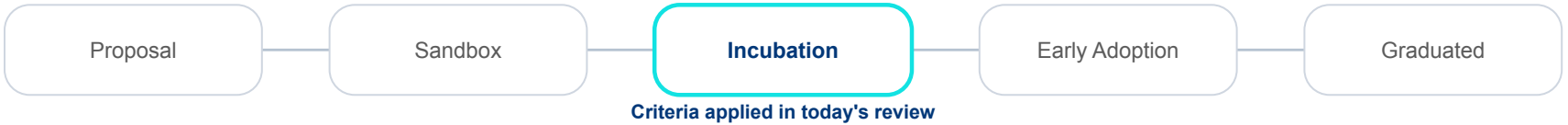
LF ENERGY

Lifecycle position

LFESS Working Group status



LF Energy project lifecycle (stages defined for contributed projects)



URPX is registered as an LFESS Working Group; the five stages above are defined for contributed projects. **How a Working Group maps to the project stages is the open question for the TAC.**

Incubation Project review criteria

Self-assessment against the Incubation criteria. URPX is registered as an LFESS working group; the TAC lifecycle stages are defined for projects.

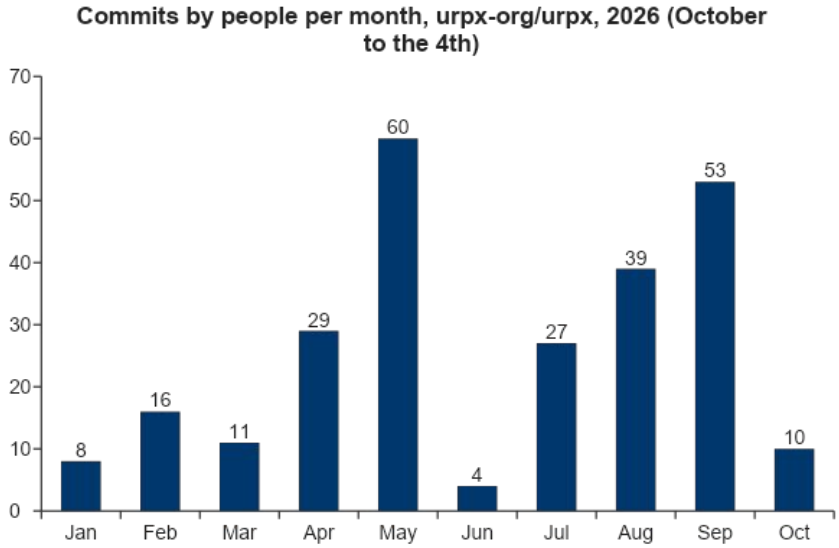
- **OpenSSF Best Practices badge: Passing.** Awarded at the public release before the meeting; the entry is filed on bestpractices.dev.
- **README in each repository: met.** README.md in urpx-org/urpx and urpx-org/urpx-wg.
- **GOVERNANCE file: met.** GOVERNANCE.md: chair, maintainers, voting committee, decision making; meetings open to anyone.
- **TSC members in LFX Project Control Center: not met.** The working group has a chair and a voting committee, not a TSC; no TSC committee is set up in LFX.
- **CODEOWNERS or COMMITTERS file: met.** .github/CODEOWNERS and MAINTAINERS.md in urpx-org/urpx; one maintainer today.

Incubation Project review criteria (continued)

- **Architecture documentation: met.** The overview page, the generated data dictionary and the diagram set under documentation/ in urpx-org/urpx.
- **Roadmap for the next year: in progress.** Working-group agendas name the candidates (highly dynamic prices at v0.6.0, emissions, an API); no roadmap document in the public repository yet.
- **Growth assessment from LFX Insights: not met.** The project is not onboarded to LFX Insights; git figures are on the Contributions slide.
- **Plan for a thriving user community: in progress.** Public release follows the IP review; co-chair, secretary and outreach roles opened at meeting #018; LF Energy blog post drafted.
- **Outline toward Early Adoption: not met.** Not written yet.

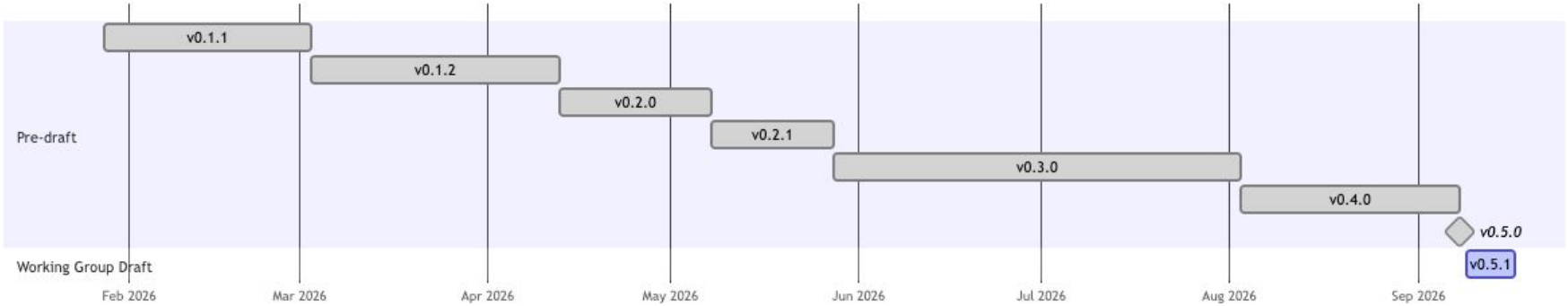
Contributions

urpx-org/urpx, the standard	
Commits by people	257
Commits by the generated-content automation	64
Contributors (people)	4
Organizations contributing commits	Flux Tailor, plus one independent contributor
Maintainers with merge rights	3
Releases tagged	8 (v0.1.1 to v0.5.1)
urpx-org/urpx-wg, the working group	
Commits	47
Contributors (people)	3
Working-group meetings held	19



Figures from git on 2026-10-05. The project is not yet onboarded to LFX Insights.

URPX Releases

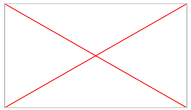


Organizations contributing and/or using in production

Contributing to the standard

Flux Tailor

Primary sponsor; chair + 2 contributors



National Laboratory of the Rockies

Collaboration partner

Supporting

Green Button Alliance

Supporter; logo used with permission

UtilityAPI

LF Energy supporter; WG participant

Using in production

- URPX is a Working Group Draft; the repository goes public after the LF Energy IP review.
- URPX serves as foundation for UtiliPub,™ Flux Tailor's rate plan publication platform.
- Further participating organizations are named in the public meeting records of urpx-org/urpx-wg.

Key Achievements in the past year

- Kickoff 2025-11-19; 19 working-group meetings, biweekly since 2026-01-14, open to anyone on the mailing list.
- Seven releases: v0.1.1 (2026-03-03), v0.1.2, v0.2.0, v0.2.1, v0.3.0 (2026-08-03), v0.4.0 and v0.5.0 (2026-09-08); v0.5.1 in preparation.
- Working Group Draft acceptance on 2026-07-17 (LFESS Sandbox to Draft).
- License transition completed 2026-09-09: W3C Document License for the specification, CDLA-Permissive-2.0 for the datasets, Apache-2.0 for code; DCO for contributions.
- Vocabulary namespace at https://urpx.org/ns/ontology/ with a published JSON-LD context; four test-case packages (twelve data files) validating against the SHACL shapes.
- Repository and website publication framework for Working Group organization: urpx (the standard), urpx-dev (development record), urpx-data-sandbox (community data); governance, maintainers, security policy and code of conduct published.
- Brand guidelines and logo package published; LF Energy project page live; Green Button Alliance support.

Repository layout

urpx-org/urpx

The standard. Ontology (OWL), SHACL shapes, the JSON-LD context, SKOS mappings, documentation, test cases, and tooling.

urpx-org/urpx-wg

Meeting agendas and minutes, the brand package, and announcements.

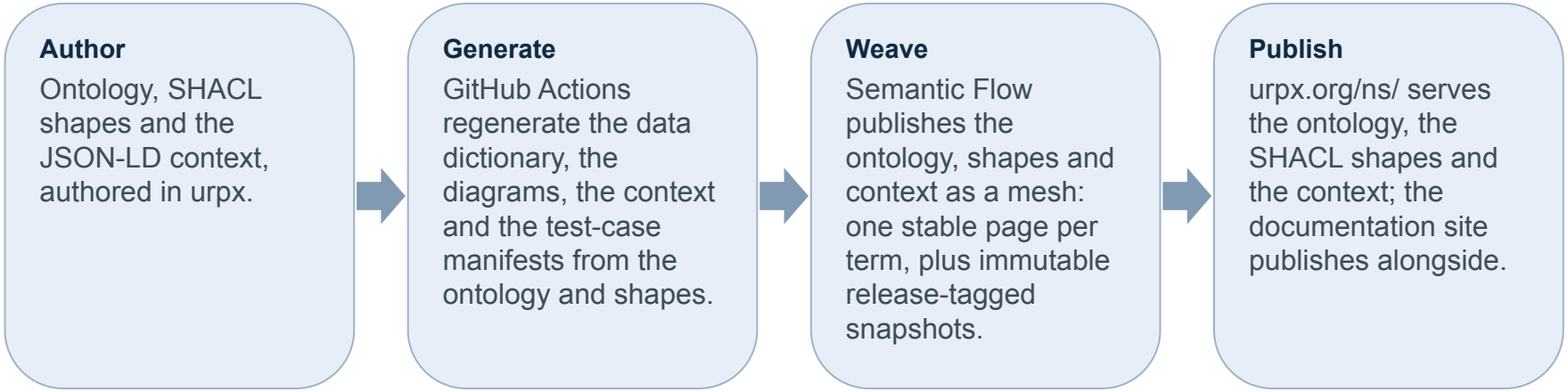
urpx-org/urpx-dev

Development record. Tasks, specifications, working reports, roadmaps, review materials, and worked sample instances.

urpx-org/urpx-data-sandbox

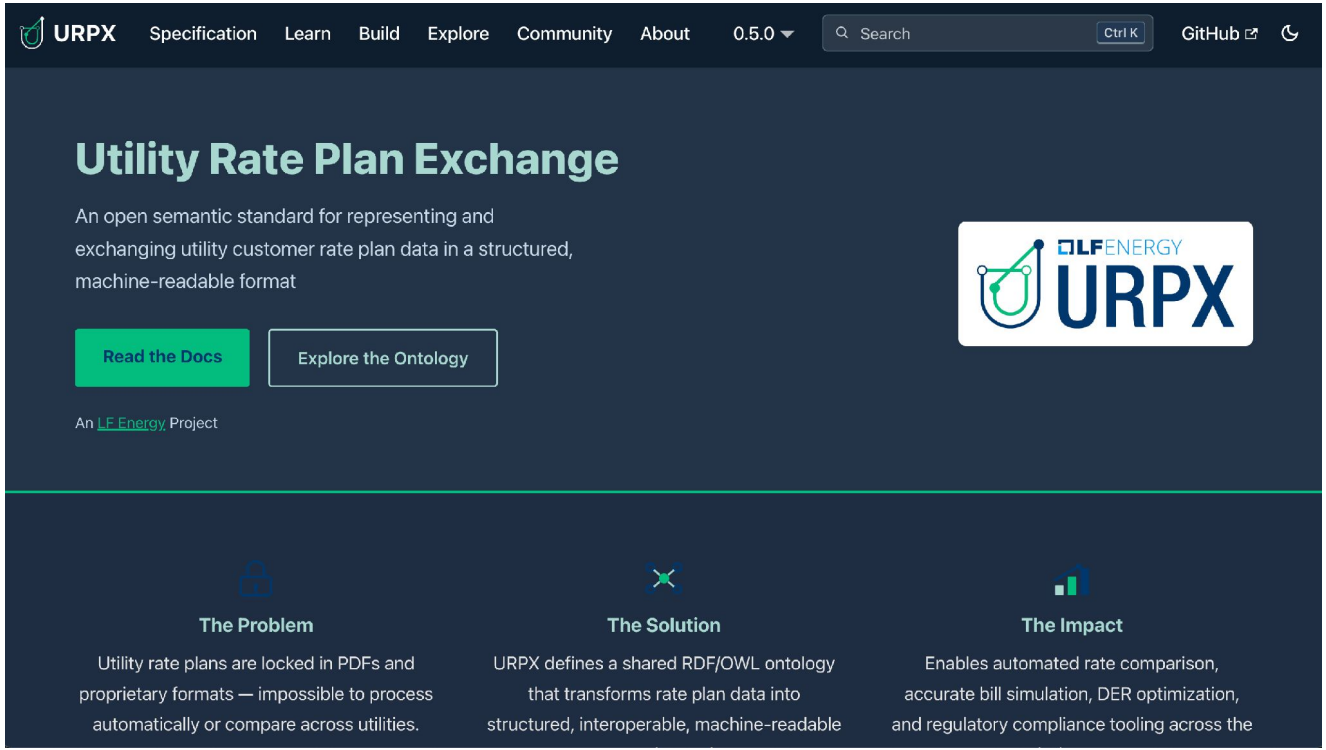
Community data. Contributed datasets for implementation and validation.

From ontology to published namespace

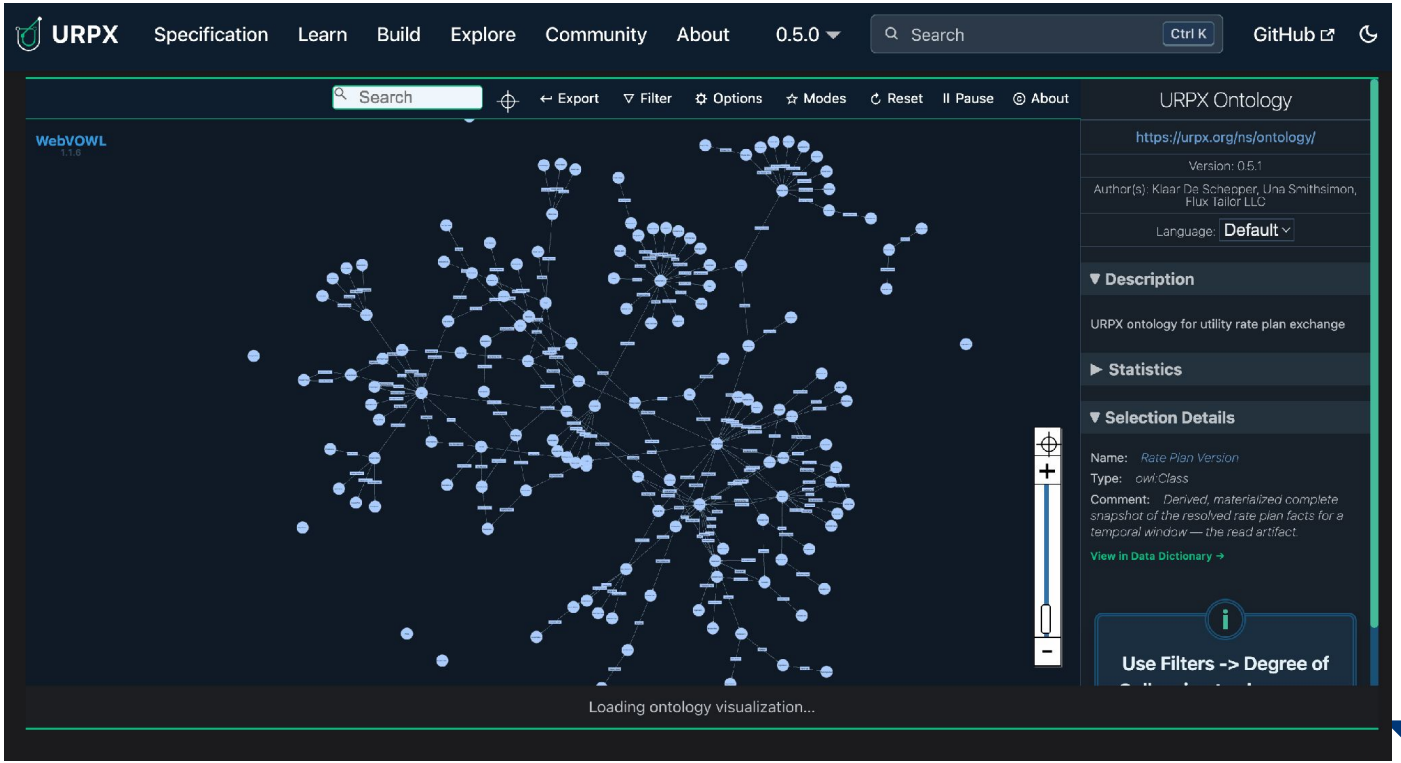


Semantic Flow (weave): github.com/semantic-flow/weave • Namespaces: urpx.org/ns/ontology/, urpx.org/ns/shacl/core/, urpx.org/ns/context

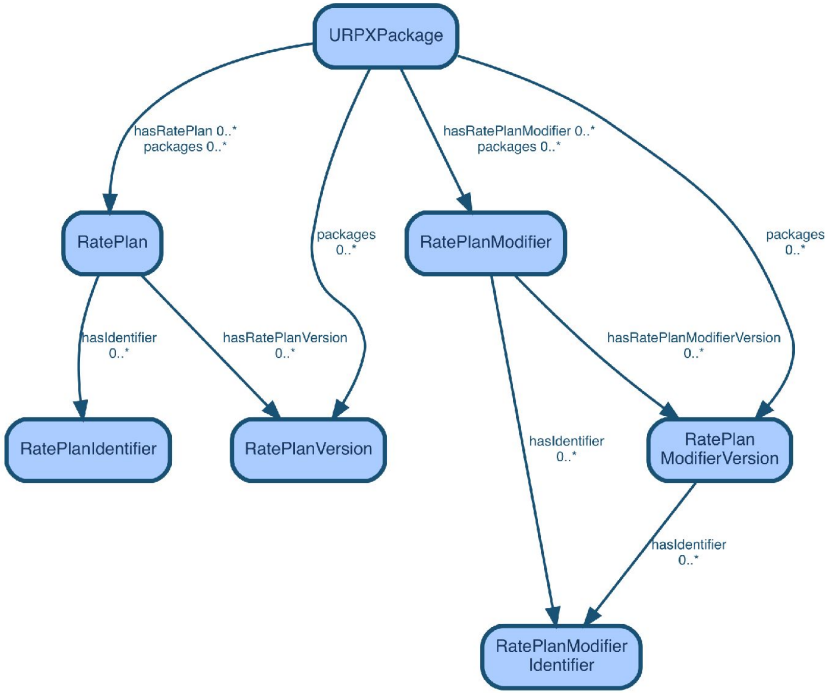
URPX.org website



Documentation: Interactive Ontology Navigator



The model at a glance: core rate plan entities



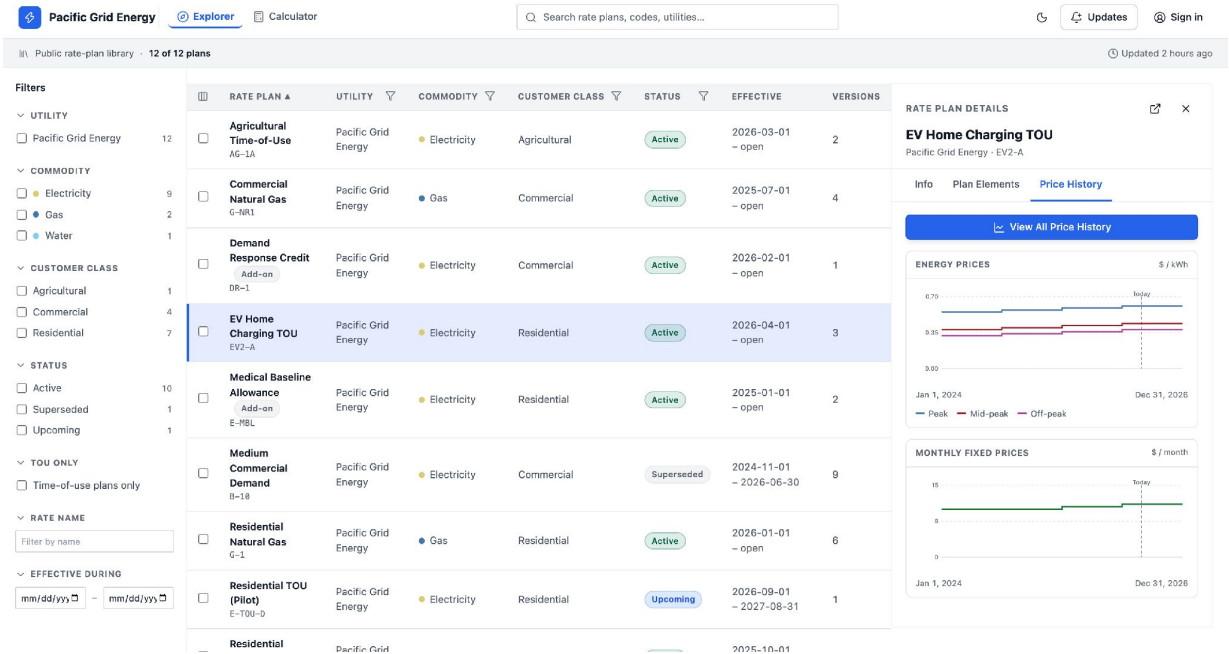
The core rate plan entities and their relationships, generated from the ontology.

Coordinating with grid and market standards

- **Standards alignment:** formal SKOS mappings from URPX to Green Button / NAESB ESPI, LF Energy CDS, CIM and OpenADR is in progress, and interoperability with IEEE 2030.5 (SEP) and Matter. URPX coordinates with usage-data, grid and demand-response standards rather than competing with them.
- **Complementary to CDS:** CDS Customer Data and CDS Registration handle customer-data access and utility discovery; URPX defines the rate plan format those customers are charged under.

URPX in production

- **Reference implementation**: UtiliPub, by Flux Tailor, is a rate plan publication platform built on URPX.



Areas the project could use help on

- **Contributors and committers:** committers from more organizations, and implementers to validate real utility data against the shapes.
- **API specification:** volunteers to help shape the common service API and a reference implementation.
- **Working-group roles:** the open co-chair and secretary roles.
- **CI/CD set up:** to ensure consistency and code quality
- **Collaborative AI-assisted workflow/harness system**
- **Launch amplification:** the LF Energy blog post and newsletter at the public release (drafts ready).

Feedback on working with LF Energy

Infrastructure in place: the GitHub organization, the urpx-wg and urpx-committee mailing lists, the LFX Zoom calendar with recordings, the artwork and landscape entries, the project page, and the brand package are all working.

LFX: a lot of surface to learn; staff support has been strong throughout.

Namespace hosting: every URPX term needs a stable, dereferenceable identifier. One harmonized hosting solution across LF Energy projects would let every standards project do this well. URPX solved it with Semantic Flow (github.com/semantic-flow/weave): one page per term, plus immutable release-tagged snapshots.

Standardization across and collaboration between standards

Marketing/PR/Events Updates

5:55 pm – 6:10 pm

Marketing Advisory Committee Chair

Chris Xie stepped down as MAC Chair, creating a vacancy.

Volunteers for chair were solicited from the current MAC membership, and only one raised their hand:

- Marco Möller, CEO, Pionix

The Linux Foundation Governing Board voted to approve the appointment.

Congratulations to Marco!

Marketing and PR Updates

Webinars

- October 27: VAREN
- November 18: EnerGNN
- December 15: OpenGridFM
- January (tentative): findings of benefit-cost framework research report
- February (tentative): TopRAO (PowSyBl)

Content

- LF Energy Expands Global Energy Ecosystem with New Members, Open Source Projects and Technical Milestones
- LF Energy Research Finds Open Source Software Can Deliver 2-5x Greater Net Value for Grid Operators
- Case study on RTE international and ENTSO-E's use of OperatorFabric in final approval stage

Project Updates

- AssetLife and Smart HEMS Benchmark webpages now live
- SEAPATH Becomes LF Energy's First Graduated Project

Upcoming Content/News Announcements

- GEISA 1.0 spec
- EU Tech Sovereignty Package position paper

Website

- New lfenergy.org launched prior to the summit

Event tracker - please review and add any additional opportunities

Use this form to submit any comms/marketing support requests

See media coverage spreadsheet or website for recent articles

Updated slide decks

- New slide template to match our website
- Updated overview deck
- Updated project slides
- To come – updated Annual Review template

Building blocks for grid analysis, visualization, and simulation

OVERVIEW

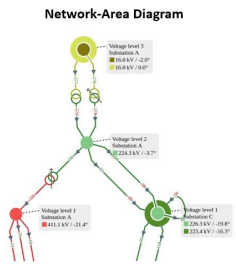
Modular, open source framework for grid modeling, analysis, and visualization, used to build and extend planning and operations tools.

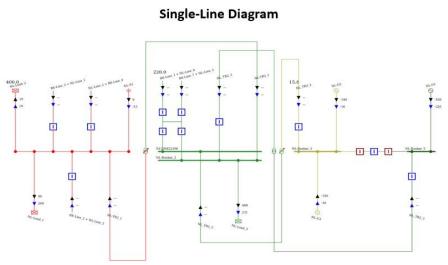
KEY CAPABILITIES

- Data exchange: CGMES, UCTE-DEF, PSS®E, PowerFactory
- Analysis: load flow, contingency, sensitivity, short-circuit
- Remedial action optimization: topology, HVDC, redispatch
- Simulation: dynamic (via Dynaωo) and time-series planning
- Network diagrams: single-line and network-area

USE CASES

- N-1 security, congestion management (TenneT case study)
- Grid reinforcement and renewable connection planning
- Capacity calculation, model merging (Baltic RCC case study)





KEY ADOPTERS & CONTRIBUTORS

Le réseau de transport d'électricité · TenneT · Baltic RCC · coreso · Artelys · grupo aia

LEARN MORE lfenergy.org/projects/powsybl

OpenGridFM Toolkit for building foundation models for power grid analysis

OVERVIEW

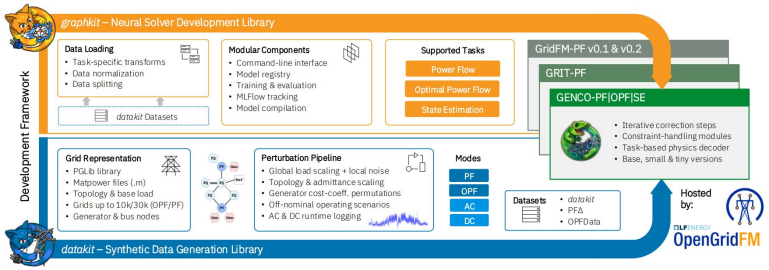
Open source toolkit for utilities and researchers to build AI models that solve power flow and related grid studies faster than conventional solvers.

KEY CAPABILITIES

- Synthetic training data: steady-state and dynamic, from test cases or a utility's own grid models
- Model training: pre-training, fine-tuning, evaluation
- Benchmarking: against AC and DC power flow baselines
- Reference model and data: GENCO models and datasets

USE CASES

- Generate datasets to train and benchmark AI models
- Fine-tune a pretrained model on a utility's own grid
- Build models for fast N-1 and N-k contingency screening



KEY ADOPTERS & CONTRIBUTORS

IBM · Hydro Québec · Artelys · Rte Le réseau de transport d'électricité · INESCTEC · Argonne NATIONAL LABORATORY

LEARN MORE lfenergy.org/projects/opengridfm

LF ENERGY SEAPATH — Real-time virtualization platform for digital substations

OVERVIEW

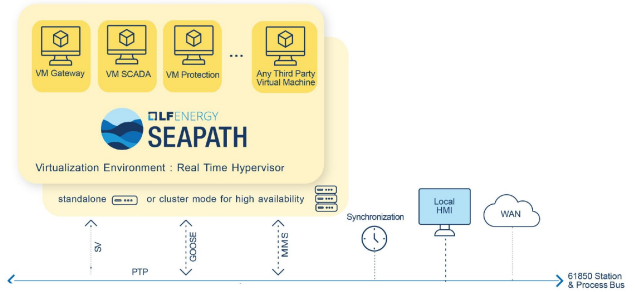
Open source, real-time virtualization platform that runs protection, automation, and control applications from multiple vendors on shared substation servers.

KEY CAPABILITIES

- Real-time virtualization: deterministic, SV latency-tested
- High availability: failover or active/active redundancy
- Multi-vendor application hosting: several vendors' VMs
- Infrastructure as code: identical setup across substations
- Cybersecurity: ANSSI BP-028 hardening, independent audit
- Time synchronization: PTP based on IEC 61850-9-3, and NTP

USE CASES

- Consolidate HMI, gateway, and automation (RTE case study)
- Run virtualized protection relays (National Grid case study)



KEY ADOPTERS & CONTRIBUTORS

RTE · Savoir-faire Linux · GE Vernova · ABB · nationalgrid · ecoPhi · elia · alliander · Red Hat · SUSE

LEARN MORE lfenergy.org/projects/seapath

OpenSTEF Machine learning pipelines for short-term energy forecasting

OVERVIEW

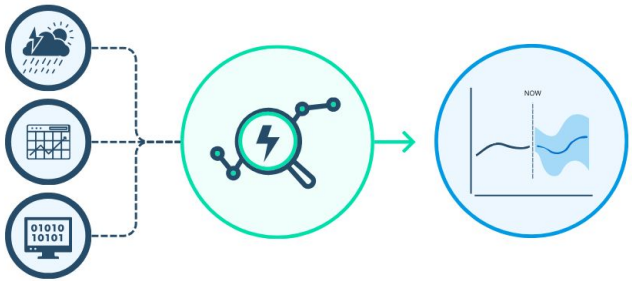
Open source Python framework for probabilistic forecasts of energy load and generation, minutes to days ahead, for grid operation and energy production planning.

KEY CAPABILITIES

- Probabilistic forecasting: minutes to days ahead with uncertainty bands
- Models: XGBoost, LightGBM, ensembles, foundation model
- Automatic retraining: models adapt as load patterns change
- Backtesting and evaluation: backtests, metrics by lead time
- Modular pipeline: swap any step; MLflow, Optuna

USE CASES

- Forecast congestion and grid losses (Alliander case study)
- Plan district heating production (Sigholm case study)



KEY ADOPTERS & CONTRIBUTORS

Alliander · Sigholm · E.ON · RTE Le réseau de transport d'électricité · SIA Partners

LEARN MORE lfenergy.org/projects/openstef

LF Energy Summit 2026 Recap

View the wrap up video

Final registration ended at 493

Photos, videos, and slides now available

Post-event survey has closed; feedback summary:

- **Overall satisfaction was high:** highest marks for keynote/breakout session quality (4.31/5), likelihood to attend a future LF Energy event (4.53/5), and likelihood to recommend the event to others (4.47/5); networking time and relationship-building also scored well (4.05 and 4.21).
- **Venue logistics were the biggest pain point:** complaints about noise in common areas, rooms and social spaces feeling too small, and unreliable WiFi
- **Session and scheduling improvements were requested:** suggestions included avoiding parallel sessions on similar topics, clearer communication of tutorial/workshop lengths, better quality control on some keynote content, and spreading the agenda across more days to reduce overlap.
- **Attendees want deeper technical and real-world content:** popular requests for future topics included more case studies and production deployment experiences, cybersecurity and digital substation content, AI's role in the energy sector, DSO-specific applications, and more hands-on/demo-based sessions with shared code.

Attendee analysis

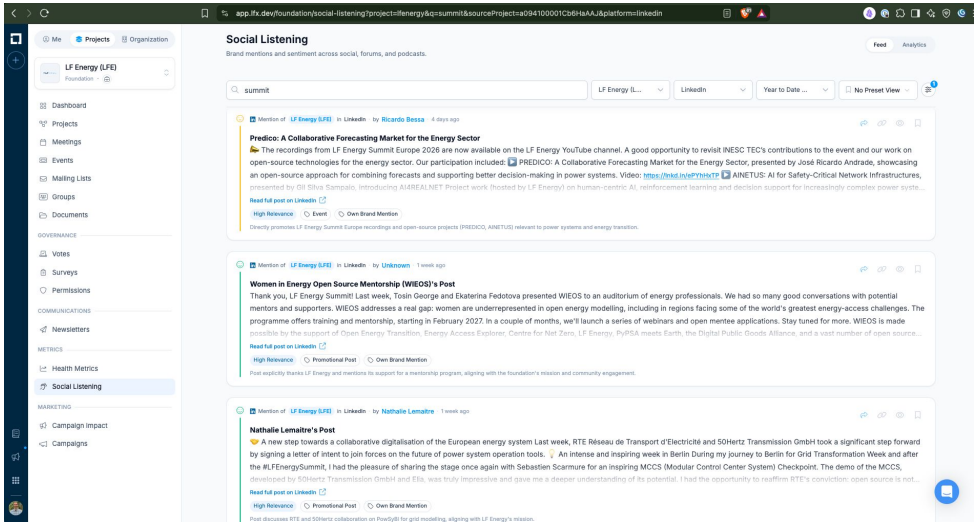
- ~40% grid operators / utilities
- ~40% vendors / consultants
- 17 TSOs
- 5 RCCs
- 13 DSOs / utilities

Type	Attendees	
Association / nonprofit / think tank	23	5%
Consultancy & services	76	15%
DSO / Utility	48	10%
Energy company (non-grid)	5	1%
Government / public sector	4	1%
Linux Foundation / event staff	12	2%
Media / analyst	3	1%
RCC	18	4%
Research / academia	58	12%
TSO	115	23%
Unclassified	22	4%
Vendor	108	22%
Total	**492**	

Type	# of Companies
Association / nonprofit / think tank	9
Consultancy & services	33
DSO / Utility	13
Energy company (non-grid)	3
Government / public sector	4
Linux Foundation / event staff	3
Media / analyst	3
RCC	5
Research / academia	36
TSO	17
Unclassified	15
Vendor	49

Social listening

- New LFX portal with improved UI and new features, including social listening
- https://app.lfx.dev/
- https://app.lfx.dev/foundation/social-listening?project=lfenergy&q=summit&sourceProject=a094100001Cb6HaAAJ&platform=linkedin



Social listening

“Open source in energy is not just the future; it is already here.”
— Open Energy Transition

“Open source is not just a technical choice for a TSO—it’s a strategic one.”
— Nathalie Lemaitre, RTE

“There’s such an openness within this community to collaborate, contribute to open source, and share work in ways that benefit the wider energy ecosystem.”
— Ava Mohammadi, Eindhoven University of Technology

“The real value was mainly in the many good conversations, new contacts and opportunities for collaboration.”
— Martijn Govers, Alliander

“It was great to see how strongly the open-source energy community is growing, with more projects and engineers collaborating across borders.”
— CRESYM

“Really enjoyed meeting people and learning more about open source projects in the energy sector!”
— Paula Passet, Sia

“So far it is a really great conference with nice talks from people who are way more expert in the energy sector than I am.”
— Andreas Prins, SUSE

“Open-source ecosystems provide us with the tools to make this work.”
— Arne Bernhardt, SOPTIM

Closing and Next Meeting

6:10 pm – 6:15 pm

Next TAC Meeting

The following meeting of the LF Energy TAC is scheduled for November 10, 2026 at 5:00 pm Central European Time. Agenda tentatively to include:

- General Updates
- Annual Review: GEISA #230
- Annual Review: AI SIG #53
- Annual Review: OSPO SIG #771
- Marketing/PR/Events update

To add agenda items, go to https://github.com/lf-energy/tac/issues/new/choose.
You can review the TAC Agenda at https://github.com/orgs/lf-energy/projects/2/views/1


LF ENERGY
Thank you!